

BDCC

Sandbox Overhaul

Technical design and implementation handoff - Version 1.0

A focused plan for making the prison sandbox feel alive through relationships, routines, consequences, conversations and player agency - while building on BDCC's existing systems.

Basis	Value
Target	A personal fork built over BDCC main, not an upstream contribution
Engine	Godot 3.x; preserve existing project conventions
Repository inspected	Alexofp/BDCC main, commit 13df5378 (2026-08-04)
Attached reference	Black Lace Stockings v4 mod structure inspected
Primary implementer	Claude Sonnet 5.5 in Visual Studio Code

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How to use this document

This is an implementation brief, not a promise to build every feature at once. Claude should use it to understand the target architecture and then implement one milestone at a time. Every milestone must compile, load old saves safely where feasible, and be play-tested before the next one begins.

Non-negotiable

Reuse BDCC's current systems first. Add new components only when the existing system cannot express the required behaviour cleanly. Keep core edits small, documented and easy to locate.

Reading order

- 1 Sections 1-4 define the design boundaries and the BDCC systems that already exist.
- 2 Sections 5-23 specify each sandbox feature in player-facing and technical terms.
- 3 Sections 24-27 define assets, migration, testing and milestone order.
- 4 Claude should not implement later milestones while foundational data is still unstable.

Definition of done for any feature

- The player can understand what happened and why.
- The feature changes at least one meaningful decision, consequence or relationship.
- It reuses shared incident, relationship, time and location data.
- It has save/load support and does not silently corrupt existing data.
- It has cooldowns or limits that prevent repetitive interruptions.
- It does not require content or assets that have not been identified.

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1. Product goal and exclusions

Goal

Transform BDCC's sandbox from a collection of loosely connected activities into a prison simulation where inmates have readable identities, routines and motives; relationships affect behaviour; the player can intervene in events; and actions create consequences that can escalate, settle or be overcome.

Target player experience

- The prison visibly continues without the player.
- NPCs approach the player for ordinary conversation as well as threats, offers and problems.
- Friendship is difficult to earn but materially useful.
- Power is useful but does not permanently delete challenge.
- Losing creates a difficult new position rather than a dead save.
- The player can usually get away with minor misconduct, while guards still feel present.
- Events emerge from existing relationships and routines instead of feeling like isolated quests.

Explicit exclusions

Do not build	Reason
Formal social-circle simulation	Cellmates, friends, gangs, lovers and owners already create sufficient links.
Full territory strategy layer	The current map is too small; use gang hangouts instead.
Mandatory hunger, sleep or hygiene survival bars	Activities should create opportunities, not chores.
Perfect off-screen simulation	Resolve distant activity cheaply and only surface meaningful results.
Automatic enemy escalation	Repeatedly defeated enemies should become deterred and stop forcing encounters.
Large base-building system	Cells need a few functional upgrades, not furniture construction.
New dialogue engine	Extend Talking.gd and ModularDialogue.
New item or clothing framework	Use Module, ItemBase, CustomInventorySlot and rigged parts.

2. Existing BDCC architecture

The following systems already provide most of the required skeleton. The overhaul should connect and extend them rather than replacing the game wholesale.

Existing system	Current role	Overhaul use
Module.gd / GlobalRegistry	Registers scenes, events, items, perks, status effects, world edits and extenders	Create one SandboxOverhaulModule and register new content through normal arrays.

Existing system	Current role	Overhaul use
InteractionSystem	Persistent pawns, locations, weighted actions and interruptions	NPC schedules, approaches, fight intervention, hangouts and guard reactions.
AloneInteraction / InteractionGoal	Chooses goals such as eat, shower, heal, work and hang out	Add sleep, go-home, job, social and gang goals; adjust scores by time and identity.
Talking.gd	Chat, flirt, attack, sex offers, restraint help and ownership offers	Refactor the chat branch into ordinary topic categories and reciprocal reactions.
ModularDialogue	Personality-sensitive forms, fillers and event adders	Generate conversation lines and contextual references; do not create a second text generator.
RelationshipSystem	Shared affection and lust plus one special relationship per NPC	Preserve compatibility, add directional feelings and derive labels such as friend or lover.
WorldHistory	Stores player-related fights, sex and pregnancy for seven days	Extend into a bounded incident ledger with witnesses, severity and knowledge.
Reputation	Level-based Alpha, Inmates, Staff and Whore stats	Register clearer reputation axes and migrate effects gradually.
Personality / FetishHolder	Six personality values and sexual interests	Drive conversation reactions, risk tolerance, combat behaviour and sexual preferences.
StatusEffect / Buff	Timed physical effects and combat modifiers	Implement injuries, after-effects and fitness condition.
World / WorldEdit	Grid rooms, pathfinding and room metadata	Add cell-block rooms and hangouts with minimal map disruption.
GameExtender	Character hooks and custom save/load data	Persist overhaul state and add time/fight hooks without scattering save code.
StageScene	Existing stand, walk, hurt, defeat, sad and other rig animations	Use existing poses for injury v1; add a custom limp animation only later.

Verified current limitations

- RelationshipEntry currently stores only shared affection and lust, so it cannot represent one-sided trust, fear or resentment.
- WorldHistory currently ignores incidents where neither participant is the player.
- The chat topic picker draws from sexual preference topics; ordinary social conversation is absent.
- Sex aftermath changes affection from physical satisfaction, which can make coercive encounters increase affection.
- GenericAttack treats surrender and post-fight punishment almost identically and has no third-party intervention path.
- Guard pawn type currently has little behaviour beyond spawn configuration.
- NPC goals already make inmates eat, shower, heal, work and wander, so schedules should extend those goals rather than replace pawn AI.

3. Extension strategy

Recommended module boundary

Create Modules/SandboxOverhaulModule/ as the home for new interactions, goals, status effects, reputation definitions, dialogue content, events, world edits and the persistent service. Keep a short file named CORE_PATCHES.md listing every edited base-game file and why.

When to extend and when to patch

Change type	Preferred method	Example
New content type already registrable	Module registration	Injury status effects, perks, items, events and world edits.
Persistent overhaul-only data	GameExtender saveLoadData	Directed feelings, incidents, gang membership and cell assignments.
New NPC behaviour	InteractionGoal, GlobalTask or PawnInteractionBase subclass	Sleep, approach player, gang hangout and group intervention.
Existing behaviour is fundamentally wrong	Small core patch plus compatibility wrapper	Sex aftermath and directed relationships.
Existing UI needs extra fields	Patch the narrow UI scene/script	NPC detail panel and relationship explanation.
New map space	WorldEdit where practical; edit map scene only if required	Additional cell-block rooms.

Avoid parallel systems

Do not leave the old affection system making decisions while a new relationship service makes different decisions. Add compatibility getters and migrate callers incrementally so there is one authoritative answer.

Core files likely to require narrow edits

- Game/InteractionSystem/Relationship/RelationshipSystem.gd
- Game/InteractionSystem/Relationship/RelationshipEntry.gd
- Game/InteractionSystem/PawnInteractionBase.gd
- Game/InteractionSystem/Interactions/Talking.gd
- Game/InteractionSystem/Interactions/GenericAttack.gd
- Game/WorldHistory/WorldHistory.gd and WHEvent.gd
- Selected NPC list/detail UI scripts

4. Shared data foundations

Most later features must read from the same small set of state. This is how the overhaul stays coherent without becoming dozens of separate simulations.

4.1 SandboxState

```
SandboxState schema_version: int npc_profiles: Dictionary[char_id -> profile] directed_relationships:
Dictionary[observer_id -> target_id -> feelings] major_memories: Dictionary[char_id -> Array[memory_ref]]
knowledge: Dictionary[knower_id -> fact_key -> fact] cell_assignments: Dictionary[char_id -> cell_id]
gang_state: Dictionary obligations: Array cooldowns: Dictionary
```

SandboxGameExtender owns and saves this object. All fields require default-safe accessors. No feature should read raw dictionary keys throughout the codebase.

4.2 One event flow

Step	Operation	Output
1	An action resolves	Fight, conversation, sex, theft, help, job failure or guard response.
2	Record an incident	Actor, target, action, severity, consent/context, location, time and outcome.
3	Determine witnesses	Participants, pawns in the room, explicit reports and later rumours.
4	Apply personal effects	Directional trust, fear, respect, desire and resentment changes.
5	Apply public effects	Only known incidents alter reputation or gang standing.
6	Create possible responses	Conversation, avoidance, retaliation, protection, reporting or reconciliation.

4.3 Complexity limits

- Keep at most 12 major memories per NPC; merge repeated similar incidents into a count and last occurrence.
- Keep minor incidents for seven days; major unresolved grievances persist until resolved or explicitly decayed.
- Simulate only spawned pawns precisely. Resolve off-screen progress once per hour or day.
- Use named helper functions for all score calculations and include debug explanations.

5. NPC identity

Player-facing behaviour

An NPC should be recognisable by appearance, combat approach, personality, reputation, sexual preferences, current goal, cell, job and affiliations. The player discovers these facts gradually.

Minimum profile

Field	Source or generation rule	Use
Combat archetype	Derived from current stats, attacks and equipment	AI action weighting and readable strengths.
Personality	Reuse PersonalityStat; expose two or three strongest descriptors	Conversation tone, risk, cruelty and patience.
Sexual preferences	Reuse LustInterests and FetishHolder	Consent, desire, flirt and sex decisions.
Current goal	One short-term goal enum	Protection, money, status, intimacy, revenge or freedom.
Cell and cellmate	Assigned by cell service	Routine, visits and home conflicts.
Job	Optional obligation assignment	Routine and income.
Gang / owner	Existing or new affiliation references	Protection, duties and inherited hostility.

Implementation

- Do not duplicate personality or fetish data in the profile.
- Calculate combat archetype on demand and cache it until stats, perks or equipment change.
- Store only new identity facts: current goal, cell, job, gang and a few generated preferences if BDCC has no source.

- Add an NPC information panel that separates Known, Suspected and Unknown facts.

Content requirement

Code can generate identity combinations. Writing is needed for readable descriptors and dialogue variants. No new visual assets are required for the first version.

6. Incident memory and knowledge

Incident record

```
{ id, type, actor_id, target_id, location_id, day, time, severity: 0..100, outcome, context, consent_context, witnesses: [char_id], public: bool, resolved: bool }
```

Knowledge rule

A character can react only if they participated, witnessed the incident, were told about it, or learned it through a controlled rumour event. This prevents omniscient reputation changes.

Reuse

Extend WorldHistory instead of creating an unrelated event history. Remove its player-only restriction, add bounded retention and add optional fields while preserving old addEvent() callers through defaults.

Not required initially

- A probabilistic rumour network
- False rumours and planted evidence
- Detailed physical evidence
- Every NPC remembering every minor conversation

7. Personal relationships

Directional feelings

Axis	Range	Meaning
Affection	-100 to +100	How much this NPC personally likes or cares about the target.
Trust	-100 to +100	Whether the target is considered dependable and safe to confide in.
Respect	-100 to +100	Assessment of competence, courage and conduct.
Fear	0 to 100	Expected personal danger from the target.
Desire	0 to 100	Sexual or romantic attraction; not affection.
Resentment	0 to 100	Unresolved grievance and desire for redress or revenge.

Compatibility with current BDCC

Current RelationshipEntry is shared by both characters. Do not simply add six shared numbers to it. Add a directional relationship store keyed by observer and target. Keep existing affection/lust getters working during migration, but route new sandbox decisions through get_feeling(observer, target, axis).

Change rules

- Ordinary pleasant conversation: usually +1 or +2 affection, sometimes +1 trust; diminishing returns.
- Meaningful help at personal cost: +5 to +15 trust or respect depending on severity.
- Broken promise: -8 to -20 trust and +5 to +15 resentment.
- Coercion: fear and resentment rise; physical pleasure must not create affection automatically.
- Winning a fair fight: respect may rise even while affection falls.
- Values are clamped and all changes include a reason string for UI and debugging.

Relationship labels

Labels are derived summaries, not separate progression bars. Friend requires affection and trust plus at least one meaningful positive memory. Best friend requires sustained history and proven reciprocal loyalty. Lover requires mutual desire, affection and trust plus an explicit commitment event.

8. Public reputation

Reputation is what the prison believes, not the player's actual stats and not one NPC's feelings. Keep the number of axes limited and definitions visible.

Axis	Scale	Primary effects
Combat reputation	0 to 100	Attack likelihood, surrender, respect and recruitment. It reflects known results, not actual power.
Defiance	-100 compliant to +100 defiant	Expected resistance to owners, gangs and guards.
Inmate standing	-100 to +100	General legitimacy and treatment by inmates.
Staff standing	-100 to +100	Guard suspicion, leniency and informant offers.
Reliability	-100 to +100	Trust in promises, jobs, gang tasks and debts.
Cruelty	0 to 100	Expectation of excessive or vindictive treatment.
Dominant notoriety	0 to 100	Known dominant behaviour; affects interested NPCs but is not universal approval.
Submissive notoriety	0 to 100	Known submissive behaviour; independent from dominant notoriety.

Implementation

- Register new RepStatBase definitions where the existing Reputation system fits.
- Combat reputation and notoriety may be non-negative even though social standing uses negative and positive values.
- Replace existing effects gradually; do not delete Alpha/Inmates/Staff/Whore before callers are migrated.
- UI must display plain-language definitions, current label, recent causes and mechanical effects.

Do not confuse Defiance and vulnerability

Defiance is a public expectation that the player will resist. Vulnerability is a temporary situation derived from injury, restraint, isolation, allies, gear and exhaustion. Vulnerability is not another reputation bar.

9. Meaningful conversations

Design goal

Conversation must be a normal part of prison life, not only a sex-preference quiz or delivery system for favours. Both the player and NPCs can initiate. Either side can respond positively, neutrally or negatively.

Conversation categories - first version

Category	Example purpose	Possible result
Check in	Ask how they are or comment on their condition	Small affection; may reveal injury, fear or mood.
Daily life	Work, meals, sleep, gym, cellmate or current activity	Familiarity and routine information.
Prison talk	Guards, searches, recent incidents or rumours	Knowledge transfer; trust gate for sensitive facts.
Personal interests	Non-sexual likes, dislikes and ambitions	Affection and identity discovery.
Relationships	Ask about owner, lover, friend, enemy or gang	Can reveal links when trust is sufficient.
Emotional response	Reassure, encourage, sympathise or challenge	Trust, respect or irritation based on personality.
Social tone	Compliment, joke, tease, flirt, insult or dismiss	Contextual affection, desire, respect or resentment.
Repair	Apologise, explain, offer compensation or request a truce	Grievance reduction if believable.

Interaction flow

- 1 Initiator selects or AI chooses a category based on social need, personality, recent memories and relationship.
- 2 ModularDialogue generates the opening line from a new form and personality-aware fillers.
- 3 The other participant chooses Warm, Honest/Neutral, Dismissive or Hostile where appropriate. Context-specific replies can replace these labels.
- 4 The initiator reacts according to personality, topic preference, relationship and context.
- 5 Apply small directed relationship changes and optionally reveal one fact.
- 6 Record a minor conversation memory and start a pair/category cooldown.

Balance and anti-grind

- The same category between the same pair gives no relationship gain again for 12 in-game hours.
- After two ordinary conversations in a day, further gains are reduced by 75 percent.
- Ordinary chat cannot create Close Friend or Best Friend without meaningful shared memories.
- NPC-initiated chat is limited by the pacing director and can be politely declined with little or no penalty.
- A hostile response may be useful role-play, but it should not always start a fight.

BDCC reuse plan

- Refactor the chat branch of Interactions/Talking.gd; preserve flirt, sex, attack and restraint actions.

- Replace addChatTopics() use of LustTopic with a new ConversationTopic registry while leaving lust topics available under a sexual-preferences category.
- Add dialogue forms to ModularDialogue/FormBanks and personality-specific lines through fillers.
- Use DialogueFillerAdder and WorldHistory to reference recent fights, pregnancy, injuries, ownership and promises.
- Use CharacterPawn.social and afterSocialInteraction() rather than adding a second social-needs meter.

Content requirement

This feature is code-light but writing-heavy. Begin with 8 categories, 4 response tones and approximately 8-12 reusable lines per personality style. Add breadth later; do not block the system on hundreds of lines.

10. Friendship and lovers

Friendship progression

Label	Minimum concept	Material benefit
Friendly	Positive affection and no major active grievance	Warmer conversation and small requests.
Friend	Affection + trust + one meaningful positive memory	Information, minor help and occasional defence.
Close friend	High affection/trust and repeated reciprocal support	Protection, storage, medical help and warnings.
Best friend	Exceptional trust and proven loyalty during serious risk	High-priority rescue, alibi, major support and proactive concern.

Repeated chat, gifts or sex can contribute but cannot alone produce best friendship. Assistance must be reciprocal; repeated exploitation reduces trust.

Lovers

- Requires an explicit commitment conversation after mutual affection, trust and desire thresholds are met.
- Exclusivity is agreed explicitly rather than assumed.
- Benefits include private conversations, visits, protection, gifts and pregnancy support.
- Jealous or obsessive behaviour requires matching personality and visible escalation; it is not a random nuisance event.

Implementation

Use derived labels for ordinary relationships. Reserve SpecialRelationshipBase for relationships requiring their own event runner, such as owner or nemesis, unless the one-special-relationship limitation is first redesigned.

11. Cells, cellmates and upgrades

Cell assignment

- Each inmate receives a cell ID and optional cellmate.
- Each cell entrance displays cell number and known occupants.

- When an NPC asks the player to visit, the cell location is automatically learned and journaled.
- Use two occupants per cell to keep blocks compact.

Cellmate behaviour

A cellmate is an ordinary NPC with extra proximity opportunities. They can talk, steal, cover for the player, store items, invite visitors, report contraband, seek protection or become a friend, lover, owner or enemy. Do not create a separate cellmate relationship score.

Upgrades

Upgrade	Effect	Risk
Lockbox	Protects money and legal items from cellmate theft	Guards can open it.
Hidden compartment	Conceals a small amount of contraband	Can be discovered during a thorough search.
Better bed	Improves fatigue and minor-injury recovery	Costs credits; may attract theft.
Medical supplies	Enables limited self-treatment	Some supplies may be contraband.
Radio	Occasional rumour or event warning	Can be confiscated.
Workbench	Repairs or modifies selected items	Tools can raise suspicion.
Entertainment	Enables social visits and small activities	Visitors increase exposure and theft risk.

Map implementation

First build one functioning block by duplicating existing cell-block visual language and room metadata. Add rooms through WorldEdit if stable; otherwise make one documented map-scene edit. Do not build all blocks before routines and assignments work.

12. Daily routines and activities

Loose schedule

Time band	Default behaviour	Exceptions
06:30-08:00	Wake, dress, leave cell, breakfast or work travel	Injury, confinement, owner order or late sleeper.
08:00-17:00	Work, activities, goals and social movement	Gang duties, medical care and personal goals.
17:00-21:00	Meals, exercise, visits, recreation and hangouts	Job shifts, owner duties or conflict.
21:00-23:00	Return to block, visit neighbours, prepare to sleep	Night activity, fear, ownership or secret meeting.
23:00-06:30	Sleep in assigned or owner-designated cell	Patrols, insomnia, confinement and major events.

The player remains free to roam. Night changes services, witnesses, guard suspicion and risk rather than normally forcing sleep.

Implementation

- Add GoalReturnToCell and GoalSleep using InteractionGoalBase.
- Modify goal scores by time of day instead of building a new scheduler.
- Use current tiredness, hunger, social and anger needs as inputs.

- Goals may be overridden by injury, ownership, gangs, obligations or current incidents.
- Randomise transition times within small windows so movement is not robotic.

Activity meaning

Activity	Direct benefit	Sandbox opportunity
Meals	Small stamina/recovery effect	Rumours, seating choices, theft, trading and gang pressure.
Gym	Maintain condition; sparring	Assess fighters, gain respect, meet training partners.
Yoga	Recovery and composure	Injury rehabilitation and calmer social group.
Showers	Clean statuses	Private conversation, visible injuries, trading and occasional risk.
Yard/recreation	Mood or recovery	Gambling, meetings, challenges and recruitment.
Fight club	Money and combat reputation	Rankings, scouting and debts.

13. NPC initiative and pacing

NPC approaches

NPCs may approach the player for ordinary chat, warnings, invitations, threats, requests, offers, apologies or demands. This should emerge through InteractionSystem interruption actions and weighted goals, not a constant random-event popup.

Pacing director - minimal version

```
pressure = unresolved_hostile_events + injury_load + confinement + urgent_obligations
if pressure is high:
    suppress new major hostile events if recent_interruption < cooldown:
        suppress non-urgent approaches
        choose at most 1 major and 2 minor player-targeted beats per 6 in-game hours
```

- Quiet time is valid and intentional.
- Ambient NPC-vs-NPC actions do not all interrupt the player.
- Repeated event types receive a longer cooldown.
- Players can dismiss ordinary conversation with minimal penalty.
- The director controls frequency only; it does not force outcomes.

Dynamic prison events

Later events such as lockdowns, inspections, shortages and gang conflict should alter schedules, prices, available rooms and encounter weights. They should reuse existing systems instead of becoming isolated minigames.

14. Combat balance and archetypes

Immediate balance action

Disable Double Down for the overhaul's first test build. It currently permits a second basic physical attack for 10 stamina and overwhelms the action economy. Do not tune all enemies around an overpowered perk.

Archetypes

Archetype	Build	AI preference	Counterplay
Bruiser	Strength + vitality	Heavy physical attacks	Exhaust, dodge or use lust.
Skirmisher	Agility	Fast attacks, dodge and stamina pressure	Accurate attacks and endurance.
Tank	Vitality	Attrition and blocking	Lust, control or stamina denial.
Seducer	Sexiness	Lust damage and distraction	High sexiness, composure and direct pressure.
Controller	Items, restraint or status actions	Disables and manipulates	Speed, preparation and resistance.
Hybrid	Balanced	Adapts without specialising	Identify current behaviour.

Rules

- Derive archetype from existing stats, attacks and equipment; do not add a second stat sheet.
- Use diminishing returns on extreme stat scaling.
- AI weights attacks according to archetype and observed weaknesses; it is not omniscient.
- Allow an Assess action and reveal clues through appearance, rumours and prior fights.
- If Double Down returns, limit it to once per fight with high stamina cost and reduced second-hit damage.

15. Fight intervention and surrender

Third-party intervention

Replace the current watch-only experience with interrupt actions available when the player is near GenericAttack or another fight interaction.

Action	Immediate effect	Social effect
Join attacker	Enter on attacker's side	Loyalty or grievance depending on relationships.
Defend target	Enter on target's side	Trust/respect for target; enemy grievance.
Break up	Use strength, respect or intimidation check	May gain respect without combat reputation.
Help escape	Create flee attempt	Trust gain; possible cowardice or guard reaction.
Call guard	Raise guard response chance	Staff standing up; possible snitch consequences.
Attack both	Create three-way hostility	Cruelty and multiple grievances.
Watch/bet	Remain outside	No forced relationship change unless support was expected.

Group combat scope

Version 1 may resolve allies as support modifiers rather than fully rendering multi-party turn combat. One ally can provide a bounded accuracy, damage, escape or morale modifier. Full multi-combatant UI is a later feature.

Surrender

- Surrender reduces expected physical injury and may soften punishment.

- It lowers public defiance and may raise submissive notoriety.
- Fighting and losing may earn respect but increases injury and resentment.
- The winner's personality and intent determine whether surrender is honoured.

Friend intervention

Friends and gangmates use the same interruption system. Their decision considers loyalty, fear, injury and expected odds. Assistance is not guaranteed simply because a label exists.

16. Injuries, fitness and treatment

Injury set

Injury	Main mechanical effect	Default duration
Bruised ribs	Reduced maximum stamina and increased pain from heavy attacks	12-36 hours
Injured arm	Lower physical damage and restraint effectiveness	18-48 hours
Injured leg	Lower agility, dodge and escape chance	18-48 hours
Concussion	Lower accuracy, assessment and resistance	12-36 hours
Severe bruising	Small global combat penalty and visible vulnerability	8-24 hours
Sexual injury	Relevant action/stamina penalties and reinjury risk	Context dependent

Implementation

- Implement each injury as a StatusEffectBase with buffs and timed recovery.
- Fight aftermath rolls injury from damage taken, closeness of fight, surrender and existing injuries.
- Doctor treatment removes time or severity in exchange for credits.
- NPC targeting can consider visible injury as temporary vulnerability.
- Cap active major injuries to avoid unreadable stacking.

Fitness

Track one condition value from 0 to 100. Base stats remain permanent; effective combat values receive a small condition modifier. Training 2-3 times per week maintains condition. Decay begins only after several inactive days and is slow.

Animation requirement

Version 1 is code-only: use existing hurt, sad, defeat or slower-walk presentation plus status icons and text. A true limp can be added by creating a new StageScene animation using the existing rig; it may require Godot animation-editor work but not necessarily newly drawn body art.

17. Guards and enforcement

Design goal

The player should get away with many offences because guards are inattentive, corrupt or absent. Guards must nevertheless influence route, timing and risk. Do not reproduce a mod where android guards intervene constantly.

Response conditions

- A guard directly witnesses the action, receives a believable report or performs a search.
- The offence severity exceeds that guard's tolerance.
- The guard is available and intervention is not clearly suicidal.
- A per-area and per-player enforcement cooldown permits the response.

Guard dispositions

Reuse PersonalityStat to derive broad behaviour: professional, lazy, cruel, corrupt or sympathetic. Do not create a large new guard personality system.

Response ladder

Level	Examples	Typical use
Ignore	Look away, warning glance	Most minor or unwitnessed behaviour.
Informal	Order clothing, stop a fight, tell player to leave	Low severity.
Search	Frisk, cell search, confiscation	Suspicion, contraband or reports.
Discipline	Fine, restraint, work penalty, stocks	Clear repeated misconduct.
Arrest	Transport, solitary or major punishment	Serious witnessed offences or resisting guards.

Winning and losing against guards

- Winning creates an immediate escape opportunity, not immunity from the institution.
- If identified, heat rises and later searches or stronger responses become more likely.
- No instant endless reinforcements. Use delayed consequences and clear warnings.
- Losing consequences depend on evidence, offence, history, surrender and guard disposition.
- Heat decays with quiet behaviour and can be reduced through bribery, service, evidence removal or serving punishment.

18. Ownership and recovery

Expanded ownership

Owner power	Player impact	Counterplay
Schedule	Curfew, bedtime or required meetings	Excuse, negotiation, escape or deliberate disobedience.
Income	Takes a wage percentage or demands payments	Hide money, change job or renegotiate.
Equipment	Requires clothing or restraints	Keys, friends, tools or medical help.
Tasks	Work, delivery, capture or service	Comply, fake completion, refuse or sabotage.
Protection	Deters some inmates and provides help	A genuine reason to accept ownership.
Punishment	Escalates for missed duties	Surrender, negotiate, gain allies or weaken owner.

Hard recovery

Recovery should come from ordinary systems, not a hidden catch-up bonus. A player who has lost money, reputation and allies may work, train, seek medical care, repair gang standing, pay for protection or voluntarily accept ownership for temporary safety.

Failure is allowed to hurt

If the player created a severe situation with no preparation or allies, recovery can take many in-game days. The requirement is a discoverable path back, not an easy path back.

Cellmate rescue example

- 1 The cellmate stops returning at night; the empty bed is the clue.
- 2 The player finds them and privately asks what happened.
- 3 The cellmate may ask for help, deny the problem, accept ownership or refuse help.
- 4 If help is wanted, the player weakens the owner's control through intimidation, combat, kidnapping, debt, keys, protection or another fitting method.
- 5 The owner retaliates according to grievance, fear and available allies.
- 6 After several successful interventions or a settlement, the cellmate resumes their routine and trust rises.

19. Gangs and hangouts

Minimal gang model

- Gang ID, name, leader, members and rank
- Primary hangout room and active time band
- Gang-to-gang relation
- Player's personal standing with each gang
- A small task list and benefit list

Do not implement territory ownership initially. A laundry gang is simply more likely to occupy the laundry and harass enemies there.

Membership

- Joining inherits the gang's active enemies and gives a modest personal-standing change with rivals.
- Leaving removes inherited hostility but does not erase personal grievances.
- Expulsion follows warning, reduced support and a final reason-based consequence where appropriate.
- Founding a gang is later content requiring reputation, loyal recruits and resources.

Tasks

Begin with four reusable templates: collect debt, intimidate target, deliver item and protect member. Capture, rescue, sabotage and spying can follow after the incident system proves stable.

20. Grievances, deterrence and reconciliation

Decision model

$\text{motivation} = \text{resentment} + \text{goal_pressure} + \text{obligation}$
 $\text{expected_cost} = \text{fear} + \text{past_failed_attempts} + \text{injury} + \text{target_allies} + \text{power_gap}$
if expected_cost clearly exceeds motivation: avoid, surrender, negotiate or stop
if motivation remains high but direct attack is poor: seek support or indirect leverage

Deterrence

- Each failed hostile attempt adds a bounded failed-attempt weight.
- Two or three decisive losses should normally stop direct attacks unless personality or obligation is extreme.
- Deterred NPCs retain resentment but no longer interrupt the player constantly.
- They can become active again if the player is badly injured, loses allies or circumstances materially change.
- The UI may state: 'Morgan has stopped openly challenging you.'

Resolution routes

Route	What it changes
Time	Minor grievances decay if no new offence occurs.
Compensation	Repay money, return property or provide equivalent value.
Apology/explanation	Works only when trust and personality make it credible.
Favour	Complete a proportionate task.
Mediation	Use owner, gang leader, guard or trusted mutual contact.
Ceasefire	Stops action without creating affection or forgiveness.
Deterrence	Fear and cost stop action while resentment remains.

21. Jobs, obligations and money

One obligation structure

```
{ source_id, owner_type, location_id, start_time, grace_minutes, duration, recurrence, reward, penalty, valid_excuses, status }
```

Use the same structure for prison jobs, owner curfews and gang tasks. This avoids three separate scheduling systems.

Valid excuses

- Stocks, restraint or confinement
- Hospital, unconsciousness or severe injury
- Guard transport or punishment
- Owner confinement
- Conflicting higher-authority obligation

Jobs

Begin by adapting mining into a generic paid-shift pattern. Add laundry, mail, cleaning or tool work later. Jobs should include occasional decisions such as stealing, finding information or helping a co-worker rather than requiring elaborate minigames.

Money uses

- Medical treatment, protection, information, bribes, equipment and clothing
- Gang dues, debt repayment, cell upgrades and restraint removal
- Hiring help that weakens a problem without deleting it automatically
- Carried money may be robbed; stored money has search or betrayal risks

22. Sex context, consent and aftermath

Required context

Dimension	Values
Consent	Willing, pressured, submitted after defeat, forced, unable
Initiator intent	Mutual pleasure, intimacy, selfish gratification, punishment, humiliation, reproduction, control
Care	Careful, normal, indifferent, reckless

After fight choices

After victory, keep Ask, Demand/Pressure and Force as distinct actions. If Ask is refused, the player may leave, negotiate or force; the later choice records the refusal and increases the social consequence.

Aftermath rules

- Physical satisfaction changes lust/desire and short-term physical effects.
- Consent, treatment and care change trust and affection.
- Coercion changes fear and resentment even if the body responded positively.
- Public coercion may change cruelty and dominant notoriety if witnessed.
- Rough consensual sex is not treated as coercive when consent and preferences support it.
- The player can request that a partner slow down or stop; success follows intent, care, relationship and leverage.

Critical core fix

Remove the current unconditional affection formula based on minimum dom/sub satisfaction in `PawnInteractionBase.doSexAftermath()`. Route aftermath through a context-aware helper. Preserve lust calculation where appropriate.

Positive outcomes

Mutually satisfying consensual encounters may grant short-lived afterglow, recovery, confidence, bonding, pillow-talk information or a favour. Use cooldowns so sex is useful but not the universal best action.

23. Pregnancy, clothing and dynamic events

Pregnancy

Keep pregnancy secondary. Add reactions from the other parent, lover, owner and close friend; optional support; medical needs; and a small number of relationship or reputation consequences. Do not create child NPCs.

Clothing

Add visual variety through BDCC's existing item pipeline. Selected clothing may provide concealment, storage, armour, staff acceptability, gang identification or attraction modifiers, but most clothing can remain cosmetic.

Attached stockings mod - reusable pattern

Mod element	What it demonstrates
Module.gd	Registers event, printing scene and ItemBase item.
BlackLaceStockings.gd	Defines visible name, description, custom slot, rigged part scene, removal text, state and price.
BlackLaceStockingsSlot.gd	Registers a CustomInventorySlot named Stockings.
BlackLaceStockings.tscn + .tres images	Provides the actual wearable visual across leg and foot variants.
Laundry event + printing scene	Adds acquisition through an existing machine scene hook.

This means new clothing mechanics are code work, while genuinely new-looking garments require rigged visual resources. The attached mod can serve as the template for additional stocking variants and a functional clothing tag interface.

Dynamic events

After the foundation is stable, add lockdown, inspection, shortage, gang conflict, tournament and corruption-investigation events. Each event temporarily modifies existing schedules, prices, guard response or room access; do not build isolated one-use systems.

24. Player information and UI

NPC record

- Identity: appearance, known personality and combat archetype
- Residence: cell, cellmate and usual hangout
- Affiliations: gang, owner, slave and lover
- Personal feelings toward player with plain-language labels
- Known preferences and current injuries
- Recent meaningful interactions and unresolved grievance causes

Feedback

Every significant relationship or reputation change includes a reason. Avoid exposing every hidden calculation, but provide enough information to make the result understandable.

Trust -12: You promised to defend Morgan but left during the fight. Fear +9: Morgan saw you defeat their owner. Affection +5: You treated Morgan's injured arm without demanding payment.

Text colours

Colour	Meaning
Green	Clearly positive or cooperative
Yellow	Uncertain, pressured or risky
Red	Openly hostile or coercive
Blue	Actionable information, location or obligation
White/grey	Neutral or unknown

Colour communicates observable tone only. A deceptive NPC can still use neutral or friendly text. Pair colour with wording or icons for accessibility.

25. Asset and animation requirements

Feature	Code only for v1?	Additional work
Relationships, reputation and incidents	Yes	UI layout changes and writing.
Meaningful conversations	Mostly	Substantial reusable dialogue writing; no art required.
Cells and schedules	Mostly	Godot map/scene editing if new rooms are added.
Cell upgrades	Yes initially	Can use text/icons; visible furniture art is optional later.
Fight intervention	Yes for support-modifier version	Full multi-character combat UI/animation is later.
Injuries	Yes for mechanics	Reuse existing hurt/sad/defeat animations and status icons.
True limping animation	No	Create/modify StageScene animation in Godot; likely reuses rig and body parts rather than new drawings.
Guards, gangs, jobs and ownership	Mostly	Dialogue/event writing and possibly UI icons.
New clothing mechanics	Yes	Existing garments can receive tags in code.
New clothing appearance	No	Rigged .tscn plus body-form/leg/foot .tres image resources, as shown by the attached mod.
New cell-block visuals	Not entirely	Existing room assets may be duplicated; bespoke backgrounds/icons are optional polish.

Recommended asset policy

- Prototype every system with existing assets first.
- Use status text and existing animations before commissioning bespoke animation.
- Do not ask Claude to fabricate binary Godot resources blindly; use the editor or clone a verified resource structure.
- For clothing, duplicate the attached proven structure and replace visual resources intentionally.

26. Save migration and testing

Schema and migration

- SandboxState stores schema_version and migrates sequentially.
- Missing directed relationships initialise from current affection/lust only as a compatibility seed, not as permanent truth.

- Existing characters receive cells, identity goals and archetypes lazily when first loaded.
- Unknown or removed character IDs are skipped safely.
- Do not delete old reputation fields until at least one migration version has shipped locally and saves have been tested.

Essential tests

Area	Acceptance test
Save/load	Create relationships, memories, cell assignment and injury; save, reload and compare.
Conversation	NPC and player initiate; all tones resolve; cooldown prevents farming; relevant facts can be learned.
Consent aftermath	Forced encounter cannot gain affection merely from satisfaction; willing good encounter can bond.
Schedules	NPC returns to assigned cell at night, sleeps and resumes goals in the morning.
Fight intervention	Player can defend friend, leave, call guard and receive correct relationship/grievance changes.
Deterrence	Enemy repeatedly defeated stops direct attacks and does not immediately re-trigger.
Guards	Minor unseen offence usually passes; witnessed serious offence may escalate without infinite interruption.
Performance	Maximum configured pawn count advances several in-game hours without runaway history growth.

Debug tools

- Inspect directed feelings with recent reasons
- List incidents known by a selected NPC
- Show current goal score breakdown
- Display retaliation motivation versus expected cost
- Force time bands, guard searches and conversation categories

27. Milestones and Claude working rules

Milestone order

Milestone	Scope	Do not include yet
0 - Baseline	Create module shell, core patch log, schema save/load and smoke test	Gameplay changes.
1 - Social foundation	Directed relationships, incident reasons, sex-aftermath fix and NPC info debug view	Gangs, cells or guards.
2 - Conversations	Ordinary topics, reciprocal responses, NPC initiation, cooldowns and ModularDialogue content	Best-friend rescue arc.
3 - Living cell block	One block, assignments, cellmates, return/sleep goals and known cell information	Multiple blocks and all upgrades.
4 - Agency in conflict	Fight intervention, surrender distinction, friendship support and deterrence	Full multi-party combat.
5 - Consequences	Injuries, fitness, doctor cost, simple guard witnessing and searches	Lockdowns and complex evidence.
6 - Economy and power	Jobs, obligations, money sinks, ownership duties and recovery routes	Large gang system.
7 - Gangs	Membership, standing, hangouts, four task templates and retaliation	Territory conquest.
8 - Expansion	Cell upgrades, lovers, pregnancy reactions, more events and clothing	Anything not proven fun.

Claude implementation rules

- 1 Before editing, inspect the named BDCC classes and identify the smallest existing extension point.
- 2 Describe planned files and core edits before applying them.
- 3 Implement only one milestone or narrowly defined slice per prompt.
- 4 Keep compatibility methods until all callers are migrated.
- 5 Use typed GDScript where consistent with the surrounding Godot 3 code.
- 6 Add save/load defaults and migration in the same change as new persistent state.
- 7 Add debug visibility for every non-trivial score.
- 8 Run parse/load checks and targeted tests after each change.
- 9 Do not invent new assets; mark asset-dependent work clearly and provide placeholders using existing resources.
- 10 Stop and report when a requested feature requires editor-authored scene or animation work that cannot be safely generated as text.

First implementation target

Recommended first slice

Build Milestone 0 and the smallest part of Milestone 1: a saved directional relationship record, reasoned change log, compatibility accessors and the correction that prevents forced sex from producing affection through satisfaction. Do not begin cells or gangs until this foundation survives save/load testing.

Success criterion for the overhaul

A normal play session should produce understandable stories without the player deliberately manufacturing every event: an NPC starts a conversation, a friendship becomes useful, a routine changes, the player notices, intervenes, creates an enemy, survives retaliation and eventually deters or resolves the conflict. Every step should arise from the same small set of relationships, incidents, locations and schedules.